Carbon Dioxide Emissions Exploratory Data Analysis

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Carbon dioxide (CO2), a colorless and odorless gas, plays a vital role in life on Earth and is a key component of the planet's natural carbon cycle. However, human activities, particularly the burning of fossil fuels, have caused a sharp rise in atmospheric CO2 levels, intensifying the greenhouse effect and contributing to climate change. Reducing CO2 emissions is essential to mitigate the harmful impacts of global warming, such as extreme weather, rising sea levels, and threats to ecosystems and human health. Shifting to renewable energy, enhancing energy efficiency, and adopting environmentally friendly practices are critical steps toward addressing climate change and securing a sustainable future for the planet.

The exploratory data analysis (EDA) revealed several noteworthy findings:

1. CO2 emissions were analyzed by country and source, with each contributing uniquely to total emissions.
2. A significant reduction in CO2 emissions occurred during the COVID-19 Pandemic.
3. Seasonal patterns were evident, especially in the Residential sector, where heating demands increased during winter months.
4. Emission sources were examined for their distribution and trends throughout the year.
5. The Power and Industries sectors were identified as the largest contributors to CO2 emissions.

Python Libraries

In [12]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
```

Dataset:

https://www.kaggle.com/datasets/saloni1712/co2-emissions

In [13]:
```
data = pd.read_csv("C:/Python Projects/Exploratory Data Analysis Project/dataset.cs

data.shape
```

Out[13]:
```
(135408, 5)
```

In [14]:
```
data.head()
```

Out[14]:

	country	date	sector	value	timestamp
0	Brazil	01/01/2019	Power	0.096799	1546300800
1	China	01/01/2019	Power	14.816100	1546300800
2	EU27 & UK	01/01/2019	Power	1.886490	1546300800
3	France	01/01/2019	Power	0.051217	1546300800
4	Germany	01/01/2019	Power	0.315002	1546300800

In [15]:
```
data.tail()
```

Out[15]:

	country	date	sector	value	timestamp
135403	Russia	31/05/2023	International Aviation	0.016524	1685491200
135404	Spain	31/05/2023	International Aviation	0.068408	1685491200
135405	UK	31/05/2023	International Aviation	0.103774	1685491200
135406	US	31/05/2023	International Aviation	0.200409	1685491200
135407	WORLD	31/05/2023	International Aviation	1.558450	1685491200

In [16]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 135408 entries, 0 to 135407
Data columns (total 5 columns):
 #   Column     Non-Null Count   Dtype  
---  ------     --------------   -----  
 0   country    135408 non-null  object 
 1   date       135408 non-null  object 
 2   sector     135408 non-null  object 
 3   value      135408 non-null  float64
 4   timestamp  135408 non-null  int64  
dtypes: float64(1), int64(1), object(3)
memory usage: 5.2+ MB
```

In [17]:
```
data.describe()
```

Out[17]:

	value	timestamp
count	135408.000000	1.354080e+05
mean	2.359439	1.615896e+09
std	5.908582	4.020589e+07
min	0.000078	1.546301e+09
25%	0.078661	1.581098e+09
50%	0.314597	1.615896e+09
75%	1.636632	1.650694e+09
max	46.263500	1.685491e+09

In [18]:

```
data.isnull().sum()
```

Out[18]:

```
country      0
date         0
sector       0
value        0
timestamp    0
dtype: int64
```

In [20]:

```
data['date'] = pd.to_datetime(data['date'],dayfirst=True)
data['Year'] = data['date'].dt.year

data
```

Out[20]:

	country	date	sector	value	timestamp	Year
0	Brazil	2019-01-01	Power	0.096799	1546300800	2019
1	China	2019-01-01	Power	14.816100	1546300800	2019
2	EU27 & UK	2019-01-01	Power	1.886490	1546300800	2019
3	France	2019-01-01	Power	0.051217	1546300800	2019
4	Germany	2019-01-01	Power	0.315002	1546300800	2019
...	...	...	...	...	...	...
135403	Russia	2023-05-31	International Aviation	0.016524	1685491200	2023
135404	Spain	2023-05-31	International Aviation	0.068408	1685491200	2023
135405	UK	2023-05-31	International Aviation	0.103774	1685491200	2023
135406	US	2023-05-31	International Aviation	0.200409	1685491200	2023
135407	WORLD	2023-05-31	International Aviation	1.558450	1685491200	2023

135408 rows × 6 columns

Exploratory Data Analysis

Comprehensive Analysis of Country-wise Carbon Dioxide Emissions Since 2019

1. Total global emission since 2019 have reached a significant level of 155k MtCO2/day
2. China, the United States, and EU27 & UK are the leading contributors.

In [21]:
```
co2_emissions_by_country = data.groupby('country').sum(numeric_only=True)['value'].

# Create the bar graph
fig = go.Figure(data=[go.Bar(x=co2_emissions_by_country['country'], y=co2_emissions

# Customize the layout
fig.update_layout(
    title="Total CO2 Emission by country since 2019",
    xaxis_title="Country",
    yaxis_title="CO2 Emissions (MtCO2/day)",
)

# Display the graph
fig.show()
```

In [22]:
```
co2_emissions_by_country = co2_emissions_by_country[~co2_emissions_by_country['coun

px.choropleth(co2_emissions_by_country, locations='country', locationmode='country
              hover_name='country', color_continuous_scale='Viridis', projection='n
```

Time Series Plot for Global Carbon Dioxide Emissions

In [23]:
```
data_world = data[data['country'] == 'WORLD']
df = data_world.groupby('date').sum(numeric_only=True)['value'].reset_index()  # Gr
df = df.sort_values(by='date')  # Sort values by date
df['Year'] = df['date'].dt.year
today = pd.to_datetime('today')
df = df[df['date'] <= today]


# Create a single graph for all years
fig = go.Figure()

# Add traces for each year to the same graph
for year in df['Year'].unique():
    df_year = df[df['Year'] == year]

    fig.add_trace(
        go.Scatter(
            x=df_year['date'],
            y=df_year['value'],
            mode='lines',
            name=str(year),
            hoverinfo='x+y',
            line_shape='linear'
```

```
        )
    )

# Update layout for the graph
fig.update_layout(
    title="Global CO2 Emission Time Series",
    xaxis_title="Date",
    yaxis_title="CO2 Emission (MtCO2/day)",
    showlegend=True,
    height=800
)

# Show the plot
fig.show()
```

2020 marks the lowest emission since 2019. The lockdown period reflects the impact of reduced economic activity on global emissions.

Time Series Plot for Carbon Dioxide Emission of Each Country

In [24]:
```
def plot_country_co2_emissions(country):
    data_country = data[data['country'] == country]
    df = data_country.groupby('date').sum(numeric_only=True)['value'].reset_index()
    df = df.sort_values(by='date')
    df['Year'] = df['date'].dt.year

    # Define a list of colors with transparency (alpha)
    colors = {
        2021: 'rgba(0, 0, 255, 0.3)',
        2022: 'rgba(0, 128, 0, 0.5)',
        2023: 'rgba(255, 0, 0, 0.6)'
    }

    fig = go.Figure()

    for year in [2021, 2022, 2023]:
        df_year = df[df['Year'] == year]

        fig.add_trace(go.Scatter(
            x=df_year['date'].dt.dayofyear,
            y=df_year['value'],
            mode='lines',
            hovertext=df_year['date'].dt.strftime('%Y-%m-%d'),  # Use the new hover
            hoverinfo='text+y',  # Show the full date and value on hover
            line_shape='linear',  # Use a linear line shape for smoother lines
            line=dict(color=colors[year]),  # Assign the color based on the year
            name=str(year)
        ))

        # Set the x-axis tick positions and labels to be the first day of each mont
        month_starts = pd.date_range(start=f'{year}-01-01', end=f'{year}-12-01', fr
        tickvals = month_starts.dayofyear
        ticktext = month_starts.strftime('%b')
        fig.update_xaxes(tickvals=tickvals, ticktext=ticktext)
```

```
    # Update the layout
    fig.update_layout(
        title= country + " CO2 Emission Time Series",
        xaxis=dict(title="Month", tickformat="%b"),  # Display only the abbreviated
        yaxis=dict(title="CO2 Emission (MtCO2/day)"),
        legend_title="Year",
        hovermode='x',
    )

    # Show the plot
    fig.show()

for country in data['country'].unique():
    plot_country_co2_emissions(country)
```

Emissions by Sector since 2019

In [25]:

```
data_world = data[data['country']=='WORLD']

co2_emissions_by_sector = data_world.groupby('sector').sum(numeric_only=True)['valu

# Create the bar graph
fig = go.Figure(data=[go.Bar(x=co2_emissions_by_sector['sector'], y=co2_emissions_b

# Customize the layout
fig.update_layout(
    title="Global CO2 Emission by sector since 2019",
    xaxis_title="Sector",
    yaxis_title="CO2 Emissions (MtCO2/day) ",
)

# Display the graph
fig.show()
```

Box Plot for Carbon Dioxide Emissions by sector since 2019

In [26]:

```
data_world = data[data['country']=='WORLD']

# Create a box plot using Plotly Express
fig = px.box(data_world, x='sector', y='value',
             title="Box Plot of Global CO2 Emissions by Sector",
             labels={'value': 'CO2 Emissions (MtCO2/day)', 'sector': 'Sector'})

# Display the graph
fig.show()
```

Time Series Plot of Global Carbon Dioxide Emissions by Sector

1. Residential Sector - display seasonality, peaking during winters with 20MtCO2/day and lows of 4.5 MtCO2/day in the summer.

2. Ground Transport - A significant decrease in Carbon dioxide emission during the lockdown period in 2020.
3. Domestic and International Aviation - Stable most of the time, but similar to Ground Transport, a significant decrease in 2020.
4. Group Transport contributes significantly more to Carbon Dioxide emission compared to Aviation combined.
5. Power sector peaks during the winter and summer times, likely due to the energy demands for heating and cooling.

In [27]:
```
# Create a dictionary to map each sector to a color with modified opacity
sector_colors = {
    'Power': 'rgba(31, 119, 180, 0.8)',             # Blue
    'Industry': 'rgba(255, 127, 14, 0.7)',          # Orange
    'Ground Transport': 'rgba(44, 160, 44, 0.6)',   # Green
    'Residential': 'rgba(214, 39, 40, 0.6)',        # Red
    'Domestic Aviation': 'rgba(148, 103, 189, 0.4)',# Purple
    'International Aviation': 'rgba(140, 86, 75, 0.3)'# Brown
}

def plot_country_and_sector_co2_emissions(fig, data_country, row):
    for sector, sector_data in data_country.groupby('sector'):
        fig.add_trace(
            go.Scatter(x=sector_data['date'], y=sector_data['value'],
                       mode='lines', line=dict(width=2, color=sector_colors[sector]
                       name=sector, showlegend=(row == 1)  # Show legend only for t
            ),
            row=row, col=1
        )

# Get a list of unique countries
unique_countries = data['country'].unique()

# Create subplots with one column and a number of rows based on the number of uniqu
fig = make_subplots(rows=len(unique_countries), cols=1,
                    subplot_titles=[country + " Time Series Plot of CO2 Emissions b

# Plot data for each country in separate subplots
for i, country in enumerate(unique_countries):
    data_country = data[data['country'] == country]
    plot_country_and_sector_co2_emissions(fig, data_country, row=i + 1)

# Update layout and display the graph
fig.update_layout(showlegend=True, height=6000, title_text="CO2 Emissions by Sector

fig.show()
```

Total Country and Sector Distribution since 2019

1. China stands as the largest contributor to Carbon Dioxide emissions globally.
2. Despite being the 2nd most populated country, India's Carbon Dioxide emission is 4th in the lead following the United States and EU27 & UK

In [28]:
```
co2_emissions_by_country = data.groupby(['country','sector']).sum(numeric_only=True
co2_emissions_by_country = co2_emissions_by_country[~co2_emissions_by_country['coun

# Create a faceted bar chart using Plotly Express
fig = px.bar(co2_emissions_by_country, x='country', y='value', color='sector', barm
             title="CO2 Emission by country & sector since 2019",
             labels={'value': 'CO2 Emissions (MtCO2/day)', 'country': 'Country','se

# Display the graph
fig.show()
```